MORPHISMS INVERTED BY THE LOCALIZATION FUNCTOR FORM THE SATURATION OF S

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5030-5085, following the statement at 5012-5028.

Editorial note. RMS2 and RMS3 for the saturation are noted 'dual to this' (5078, 5084); they are literal arrow-reversal mirrors of the executed LMS2 and LMS3 arguments.

Allowed standard prerequisites (editorial).

- Axioms MS1-MS4 of a multiplicative system. Definition of a multiplicative system in this chapter (MS2 = LMS2 + RMS2, MS3 = LMS3 + RMS3). The closure properties verified for the saturation; the proof checks LMS2, LMS3 and notes the RMS halves dual.
- Both-sided fraction descriptions. Lemma lemma-multiplicative-system of this chapter, invoked at proof lines 5037-5040. Allows writing the inverse of Q(f) both as a left fraction s^{-1}g and a right fraction h t^{-1}.

Source reference. Original lemma 04VL: sources/stacks-categories/pinned-original-source.tex, source lines 4978–4978. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a category and let S be a multiplicative system. Denote $Q : \mathcal{C} \to S^{-1}\mathcal{C}$ the localization functor. The set*

$$\hat{S} = \{f \in \mathit{Arrows}(\mathcal{C}) \mid Q(f) \text{ is an isomorphism}\}$$

is equal to

$$S' = \{f \in \mathit{Arrows}(\mathcal{C}) \mid \text{there exist } g, h \text{ such that } gf, fh \in S\}$$

and is the smallest saturated multiplicative system containing S. In particular, if S is saturated, then $\hat{S} = S$.

Proof. It is clear that $S \subset S' \subset \hat{S}$ because elements of S' map to morphisms in $S^{-1}\mathcal{C}$ which have both left and right inverses. Note that S' satisfies MS4, and that $\hat{S}$ satisfies MS1. Next, we prove that $S' = \hat{S}$.

Let $f \in \hat{S}$. Let $s^{-1}g = ht^{-1}$ be the inverse morphism in $S^{-1}\mathcal{C}$. (We may use both left fractions and right fractions to describe morphisms in $S^{-1}\mathcal{C}$, see Lemma 04VL.)

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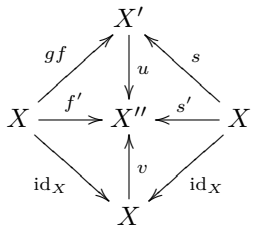
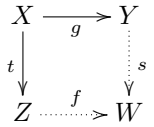
The relation $\mathrm{id}_X = s^{-1}gf$ in $S^{-1}\mathcal{C}$ means there exists a commutative diagram

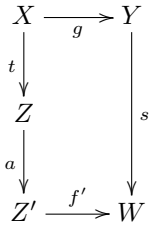


for some morphisms f', u, v and $s' \in S$. Hence $ugf = s' \in S$. Similarly, using that $\mathrm{id}_Y = fht^{-1}$ one proves that $fhw \in S$ for some w. We conclude that $f \in S'$. Thus $S' = \hat{S}$. Provided we prove that $S' = \hat{S}$ is a multiplicative system it is now clear that this implies that $S' = \hat{S}$ is the smallest saturated system containing S.

Our remarks above take care of MS1 and MS4, so to finish the proof of the lemma we have to show that LMS2, RMS2, LMS3, RMS3 hold for $\hat{S}$. Let us check that LMS2 holds for $\hat{S}$. Suppose we have a solid diagram



with $t \in \hat{S}$. Pick a morphism $a : Z \to Z'$ such that $at \in S$. Then we can use LMS2 for S to find a commutative diagram



and setting $f = f' \circ a$ we win. The proof of RMS2 is dual to this. Finally, suppose given a pair of morphisms $f, g : X \to Y$ and $t \in \hat{S}$ with target X such that $ft = gt$. Then we pick a morphism b such that $tb \in S$. Then $ftb = gtb$ which implies by LMS3 for S that there exists an $s \in S$ with source Y such that $sf = sg$ as desired. The proof of RMS3 is dual to this. $\square$